On Distinguishing Trees by Their Chromatic Symmetric Functions:
A Report on Martin–Morin–Wagner

Math 725 Independent Project

Abstract

This report is an exposition of, and a small extension to, the paper *On Distinguishing Trees by Their Chromatic Symmetric Functions* by J. L. Martin, M. Morin, and J. D. Wagner [6]. The central question, raised by Stanley [8], is whether the chromatic symmetric function $\mathbf{X}_T$ is a complete isomorphism invariant on the class of trees. After developing the necessary background, we present a complete proof, in our own words, of the paper's caterpillar reconstruction theorem (Theorem 5.1), which shows that caterpillars with strictly positive and pairwise distinct leaf numbers are determined up to isomorphism by their chromatic symmetric functions. We then work out two examples in detail and propose a strengthening of the paper's main open question, supported by computational evidence and a partial result.

1 Introduction

A central theme in algebraic graph theory is the search for combinatorial invariants that distinguish graphs up to isomorphism. The *chromatic polynomial* $\chi_G(k)$, which counts proper colorings of G with at most k colors, is well known but extremely weak on trees: every tree on n vertices satisfies $\chi_T(k) = k(k-1)^{n-1}$, so the chromatic polynomial cannot distinguish two non-isomorphic trees. In 1995, Stanley [8] introduced a much finer invariant.

Definition 1.1. Let $G = (V, E)$ be a finite simple graph. The *chromatic symmetric function* of G is

$$\mathbf{X}_G(x_1, x_2, \ldots) \;=\; \sum_{\kappa} \prod_{v \in V} x_{\kappa(v)},$$

where the sum runs over all proper colorings $\kappa\colon V \to \mathbb{P}$ of G by positive integers. Permuting the variables permutes the colorings, so $\mathbf{X}_G$ is a symmetric function, homogeneous of degree $|V|$.

The chromatic polynomial is recovered from $\mathbf{X}_G$ by setting $x_1 = \cdots = x_k = 1$ and $x_i = 0$ for $i > k$. Thus $\mathbf{X}_G$ refines the chromatic polynomial. Stanley showed that $\mathbf{X}_G$ is *not* a complete isomorphism invariant on graphs in general (i.e., there exist two non-isomorphic graphs that share the same value of $\mathbf{X}_G$), exhibiting two non-isomorphic five-vertex graphs with the same chromatic symmetric function [8]. Whether $\mathbf{X}_T$ is a complete invariant on the restricted class of *trees*—meaning that no two non-isomorphic trees can share the same chromatic symmetric function—remains open and is one of the most prominent unsolved problems in algebraic combinatorics. As of this writing, computational verification due to Tan [10] confirms the conjecture for trees on at most 23 vertices.

The paper of Martin, Morin, and Wagner [6] attacks this question from two complementary angles. First, they show that $\mathbf{X}_T$ is at least as strong an invariant as the *subtree polynomial* $\mathbf{S}_T(q, r)$, which is the bivariate generating function counting subtrees of T by their numbers of edges and leaves. In particular, the path and degree sequences of T can be extracted from $\mathbf{X}_T$.

Second, they prove that $\mathbf{X}_T$ *does* reconstruct T when T is restricted to certain natural sub-classes, including all spiders, several families of caterpillars, and (in the unicyclic setting) all squids.

The remainder of this report is organized as follows. Section 2 fixes notation and reviews symmetric-function preliminaries. Section 3 states Stanley's power-sum expansion of $\mathbf{X}_G$ and explains why it is especially well-behaved on trees. Section 4 catalogs the basic graph invariants recoverable from $\mathbf{X}_T$. Section 5 contains the main theorem of this report—a caterpillar reconstruction theorem due to Martin–Morin–Wagner—together with a complete, self-contained proof in our own words. Section 6 works out two original examples. Section 7 states a conjecture, supported by partial results, and Section 8 comments on the accompanying SageMath implementation. Appendix A summarizes the changes made between the draft and final submission in response to peer review.

2 Preliminaries on symmetric functions and trees

We follow [5, 9] for symmetric functions and [1] for graphs.

Partitions. A *partition* of n is a weakly decreasing sequence $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_\ell)$ of positive integers summing to n. We write $\lambda \vdash n$. The number $\ell = \ell(\lambda)$ is the *length* of λ. We say λ is *singleton-free* if every part is ≥ 2.

Power-sum symmetric functions. For each $k \in \mathbb{P}$ define $p_k = \sum_{i\geq 1} x_i^k$, and for a partition λ set $p_\lambda = \prod_{j=1}^{\ell(\lambda)} p_{\lambda_j}$. The set $\{p_\lambda : \lambda \vdash n\}$ is a $\mathbb{Q}$-vector-space basis for the space Λ^n of homogeneous symmetric functions of degree n.

Trees. A *tree* $T = (V, E)$ is a connected graph with $|E| = |V| - 1$; equivalently, a connected acyclic graph. A *leaf* is a vertex of degree 1, and the unique edge incident to a leaf is a *leaf edge*. We write $\delta_j(T)$ for the number of degree-j vertices of T, and $\pi_i(T)$ for the number of paths in T with exactly i edges.

Caterpillars. A *caterpillar* is a tree whose non-leaf vertices induce a nontrivial path, called the *spine*. We label the spine as $v_0, v_1, \ldots, v_s$ (so it has $s+1$ vertices and s edges), and denote by f_i the number of leaves attached to v_i, called the *i-th leaf number* of T. Note that $f_0, f_s \geq 1$ (otherwise v_0 or v_s would itself be a leaf and not lie on the spine), but f_i for $0 < i < s$ may be zero.

3 Stanley's expansion and tree-specific structure

The starting point of [6] is the following expansion of $\mathbf{X}_G$ due to Stanley [8, Thm. 2.5].

Theorem 3.1 (Stanley's expansion)**.** *For any finite graph $G = (V, E)$,*

$$\mathbf{X}_G \;=\; \sum_{A\subseteq E} (-1)^{|A|}\, p_{\mathrm{type}(A)}, \tag{1}$$

where $\mathrm{type}(A)$ *is the partition whose parts are the orders of the connected components of the spanning subgraph* (V, A)*.*

We write $c_\lambda(G)$ for the coefficient of p_λ in the power-sum expansion of $\mathbf{X}_G$, so that $\mathbf{X}_G = \sum_{\lambda \vdash n} c_\lambda(G)\, p_\lambda$.

When $G = T$ is a tree, the expansion (1) simplifies dramatically. The reason is that every $A \subseteq E(T)$ is necessarily acyclic, so the spanning subgraph (V, A) has exactly $n - |A|$ connected components. This means $\ell(\mathrm{type}(A)) = n - |A|$, and consequently the length of $\mathrm{type}(A)$ alone determines $|A|$ and hence the sign $(-1)^{|A|}$. In particular, all edge subsets A with the same type λ contribute the same sign $(-1)^{n-\ell(\lambda)}$ to $c_\lambda(T)$, and we obtain the clean formula

$$c_\lambda(T) \;=\; (-1)^{n-\ell(\lambda)} \cdot \big|\{A \subseteq E(T) : \mathrm{type}(A) = \lambda\}\big|. \tag{2}$$

Equation (2) is the combinatorial linchpin of the entire paper, and we shall use it repeatedly.

4 Recovering classical invariants

We summarize the basic invariants of a graph (or tree) that are captured by $\mathbf{X}_G$, following [6, Prop. 2 & Cor. 5].

Proposition 4.1. *For any graph $G = (V, E)$, the chromatic symmetric function $\mathbf{X}_G$ determines:*

(i) *the number of vertices, $n = |V|$;*

(ii) *the number of edges, $|E| = -c_{(2,1,1,\ldots,1)}(G)$;*

(iii) *the number of connected components,* $\min\{\ell(\lambda) : c_\lambda(G) \neq 0\}$.

If T is a tree, then $\mathbf{X}_T$ also determines:

(iv) *the number of j-vertex subtrees, namely $\big|c_{(j,1,\ldots,1)}(T)\big|$ for each $j \geq 2$;*

(v) *the number of leaves of T, namely $|c_{(n-1,1)}(T)|$;*

(vi) *the path sequence $(\pi_i(T))_{i\geq 1}$ and the degree sequence $(\delta_j(T))_{j\geq 1}$.*

The recovery of the path and degree sequences is in fact a corollary of the much stronger result of Martin–Morin–Wagner that the *subtree polynomial* $\mathbf{S}_T(q, r) = \sum_S q^{|E(S)|}\, r^{|L(S)|}$, where the sum is over edge-nonempty subtrees S of T and $L(S)$ denotes the leaf edges of S, can be extracted from $\mathbf{X}_T$ by an explicit linear formula [6, Thm. 1]. The path sequence appears as the coefficients of $q^i r^2$ in $\mathbf{S}_T$, and the star sequence (equivalent to the degree sequence) as the coefficients of $q^k r^k$.

5 The main theorem: caterpillars with distinct leaf numbers

The principal result we shall prove is the caterpillar reconstruction theorem of [6, Thm. 11]. It illustrates beautifully how the combinatorial interpretation (2) of $c_\lambda(T)$ on trees converts a structural question—"which tree is T?"—into the combinatorial problem of identifying which edge-subsets realize which types.

Theorem 5.1 (Martin–Morin–Wagner)**.** *Let T be a caterpillar with spine $v_0, v_1, \ldots, v_s$ and leaf numbers $f_0, f_1, \ldots, f_s$. Suppose every f_i is strictly positive and that the f_i are pairwise distinct. Then T is determined up to isomorphism by its chromatic symmetric function $\mathbf{X}_T$.*

Proof. Fix the caterpillar T as in the statement, and set $n = |V(T)| = (s+1) + \sum_{i=0}^{s} f_i$. Let $L \subseteq E(T)$ denote the set of leaf edges of T, and write $E_{\text{spine}} = \{e_1, \ldots, e_s\}$ for the spine edges, where $e_i = v_{i-1}v_i$. Throughout, partitions are equated with their multisets of parts.

We will show that the data

$$\mathcal{D}(T) \;:=\; \big\{c_\lambda(T) : \lambda \vdash n\big\}$$

suffices to recover $(f_0, f_1, \ldots, f_s)$ up to reversal. The isomorphism type of T depends only on this sequence up to reversal (reflecting the spine end-to-end produces an isomorphic caterpillar), so this is enough.

Step 1: Edge subsets with singleton-free type are exactly the supersets of L. Suppose $A \subseteq E(T)$ has type(A) singleton-free, i.e. every component of the spanning subgraph (V, A) has order ≥ 2. In particular, every leaf w of T lies in a component of size ≥ 2. Since the unique neighbor of a leaf is its sole neighbor in T, the only way for w to belong to a component of size ≥ 2 in (V, A) is for the leaf edge incident to w to lie in A; otherwise w is isolated in (V, A) and forms a singleton component, contradicting our assumption. This forces $L \subseteq A$.

Conversely, if $A \supseteq L$, then for every spine vertex v_i the component of (V, A) containing v_i contains at least the $f_i \geq 1$ leaves attached to v_i together with v_i itself, so it has order $\geq f_i + 1 \geq 2$. Every leaf is also in such a component (the same one as its spine neighbor), so every component has order ≥ 2 and type(A) is singleton-free.

Consequently, edge sets A contributing to a singleton-free $c_\lambda(T)$ are precisely those of the form $A = L \cup B$ with $B \subseteq E_{\text{spine}}$.

Step 2: Identifying $\lambda :=$ type(L). With $A = L \cup B$, removing the $|B|$ edges of B from $L \cup E_{\text{spine}} = E(T)$ partitions T into $|B| + 1$ subtrees, hence merges components of (V, L) as follows: the $s+1$ components $\{v_i\} \cup \{\text{leaves at } v_i\}$ of (V, L) are joined into exactly $(s+1) - |B|$ groups by the spine edges in B. Thus $\ell(\text{type}(L \cup B)) = (s+1) - |B|$, and the maximum length is achieved exactly when $B = \emptyset$.

Set $\lambda :=$ type(L). Then λ has parts $f_0 + 1, f_1 + 1, \ldots, f_s + 1$ (one for each spine vertex). Step 1 and the preceding paragraph show:

$$\lambda \text{ is the } \textit{unique} \text{ singleton-free partition of } n \text{ of length } s+1 \text{ with } c_\lambda(T) \neq 0,$$

and indeed $c_\lambda(T) = (-1)^{n-(s+1)}$ since exactly one edge set ($A = L$ itself) has type λ. We can read off $s+1$ as the maximum length of a singleton-free partition appearing in $\mathbf{X}_T$ with nonzero coefficient, and then λ as the unique such partition of that length. In particular, the multiset $\{f_0 + 1, f_1 + 1, \ldots, f_s + 1\}$, hence the multiset $\{f_0, f_1, \ldots, f_s\}$, is determined by $\mathbf{X}_T$.

Note that since the f_i are pairwise distinct, all parts of λ are pairwise distinct.

Step 3: Identifying the partitions $\mu_i :=$ type$(L \cup \{e_i\})$. Now consider $|B| = 1$, say $B = \{e_i\}$. The edge $e_i = v_{i-1}v_i$ merges precisely the two components of (V, L) containing v_{i-1} and v_i, of sizes $f_{i-1} + 1$ and $f_i + 1$, into a single component of size $f_{i-1} + f_i + 2$; the other $s - 1$ components are unaffected. Hence

$$\mu_i \;:=\; \text{type}(L \cup \{e_i\}) \;=\; \big(\lambda \setminus \{f_{i-1} + 1,\ f_i + 1\}\big) \cup \{f_{i-1} + f_i + 2\},$$

where we manipulate λ as a multiset of parts.

Claim: the partitions $\mu_1, \ldots, \mu_s$ are pairwise distinct. Suppose $\mu_i = \mu_j$ as multisets, with $i \neq j$. Comparing multisets, the symmetric difference $\mu_i \triangle \lambda$ equals $\{f_{i-1} + 1,\ f_i + 1,\ f_{i-1} + f_i + 2\}$, and similarly for μ_j. The element $f_{i-1} + f_i + 2$ strictly exceeds both $f_{i-1} + 1$ and $f_i + 1$ (because each $f_k \geq 1$, hence $f_{i-1} + f_i + 2 \geq f_{i-1} + 1 + f_i \geq \max(f_{i-1}, f_i) + 2$). Hence in either symmetric

difference, the merged part is the unique maximum. Equating the symmetric differences forces $f_{i-1}+f_i+2 = f_{j-1}+f_j+2$ and $\{f_{i-1}+1,\ f_i+1\} = \{f_{j-1}+1,\ f_j+1\}$, i.e. $\{f_{i-1}, f_i\} = \{f_{j-1}, f_j\}$. Since the f_k are pairwise distinct, this forces $\{i-1, i\} = \{j-1, j\}$, hence $i = j$. This contradicts our assumption, proving the claim.

By Step 1 and the length count from Step 2, the edge subsets with type singleton-free of length exactly s are precisely $L \cup \{e_1\}, \ldots, L \cup \{e_s\}$. By the claim, these contribute to s *distinct* coefficients $c_{\mu_1}(T), \ldots, c_{\mu_s}(T)$, each equal to $(-1)^{n-s}$. Therefore the multiset $\{\mu_1, \ldots, \mu_s\}$ is exactly the set of singleton-free partitions μ of n with $\ell(\mu) = s$ and $c_\mu(T) \neq 0$, and is likewise determined by $\mathbf{X}_T$.

Step 4: Reconstructing the leaf-number sequence. Because the parts of λ are pairwise distinct, for each μ_i there is a uniquely determined unordered pair $\{a_i, b_i\}$ of parts of λ that are absent from μ_i, and a uniquely determined "new" part $a_i + b_i$ in μ_i. Step 3 shows that $\{a_i, b_i\} = \{f_{i-1}+1,\ f_i+1\}$, equivalently $\{a_i - 1,\ b_i - 1\} = \{f_{i-1},\ f_i\}$.

Form an auxiliary graph H on the vertex set $V(H) = \{f_0, f_1, \ldots, f_s\}$ (a set of $s+1$ *distinct* non-negative integers, by hypothesis) with edges $E(H) = \{\{f_{i-1}, f_i\} : 1 \leq i \leq s\}$. Steps 2 and 3 show that $V(H)$ and $E(H)$ are determined by $\mathbf{X}_T$. But H is the path $f_0 – f_1 – \cdots – f_s$ on $s+1$ vertices, and a path on labeled vertices is determined up to reversal by its edge set. Hence the sequence $(f_0, \ldots, f_s)$ is determined by $\mathbf{X}_T$ up to reversal, which (as noted above) determines the caterpillar T up to isomorphism. □

Remark 5.2. The hypothesis that the f_i are pairwise distinct can be weakened. The argument in Step 3 only used distinctness to ensure that whenever $\mu_i = \mu_j$ one has $\{f_{i-1}, f_i\} = \{f_{j-1}, f_j\}$. If we instead assume that for each value k, the set $\{i : f_i = k\}$ forms a contiguous subpath of indices, then the argument still allows us to identify which adjacencies $\{f_{i-1}, f_i\}$ appear as multisets, and hence to reconstruct T. This relaxation is mentioned in passing in [6, §4].

6 Worked examples

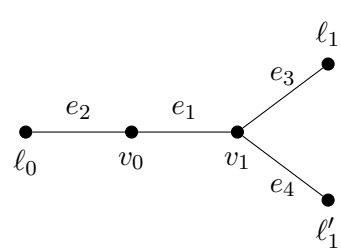
We work two examples that do not appear in [6]. The first explicitly computes the chromatic symmetric function of a small tree and verifies the identities of Proposition 4.1. The second illustrates Theorem 5.1 on a caterpillar with three spine vertices.

6.1 Example: full power-sum expansion of a five-vertex tree

Let T be the tree on $n = 5$ vertices with vertex set $\{v_0, v_1, \ell_0, \ell_1, \ell_1'\}$ and edge set $E(T) = \{e_1, e_2, e_3, e_4\}$, where

$$e_1 = v_0 v_1, \quad e_2 = v_0 \ell_0, \quad e_3 = v_1 \ell_1, \quad e_4 = v_1 \ell_1'.$$

Equivalently, T is the caterpillar with spine v_0, v_1 and leaf numbers $f_0 = 1$, $f_1 = 2$, or equivalently the spider $T_{(2,1,1)}$. A picture is shown below.



We compute $\mathbf{X}_T = \sum_{A \subseteq E(T)} (-1)^{|A|} p_{\mathrm{type}(A)}$ by enumerating all $2^4 = 16$ edge subsets $A \subseteq E(T)$ and tabulating their types. In the table below, the rows are organized by $|A|$; each row groups the subsets that share the same type, lists their component sizes, and records the resulting partition type(A). The total signed contribution to $c_\lambda(T)$ is then read off as $(-1)^{|A|}$ times the row's multiplicity.

$\lvert A\rvert$	subsets	component sizes	type
0	$\varnothing$	five singletons	(1^5)
1	each of $\{e_1\}, \{e_2\}, \{e_3\}, \{e_4\}$	one edge plus 3 isolated	$(2,1,1,1)$
2	$\{e_1,e_2\}, \{e_1,e_3\}, \{e_1,e_4\}, \{e_3,e_4\}$	one P_3 plus 2 isolated	$(3,1,1)$
2	$\{e_2,e_3\}, \{e_2,e_4\}$	two edges plus 1 isolated	$(2,2,1)$
3	$\{e_1,e_2,e_3\}, \{e_1,e_2,e_4\}, \{e_1,e_3,e_4\}$	one P_4/star plus 1	$(4,1)$
3	$\{e_2,e_3,e_4\}$	disjoint P_2 and P_3	$(3,2)$
4	$E(T)$	all of T	(5)

Tallying the signed contributions to each $c_\lambda(T)$ gives

$$\begin{aligned}\mathbf{X}_T \;=\; & p_{(1,1,1,1,1)} \;-\; 4\,p_{(2,1,1,1)} \;+\; 4\,p_{(3,1,1)} \;+\; 2\,p_{(2,2,1)} \\ & -\; 3\,p_{(4,1)} \;-\; p_{(3,2)} \;+\; p_{(5)}.\end{aligned} \tag{3}$$

We now verify each part of Proposition 4.1:

(i) Every monomial of every p_λ in (3) has total degree $5 = |V(T)|$.

(ii) $-c_{(2,1,1,1)}(T) = -(-4) = 4 = |E(T)|$.

(iii) The minimum length of a partition with non-zero coefficient is $\ell((5)) = 1$, matching the 1 component of T.

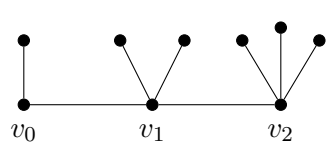
(iv) For $j = 3$ we have $|c_{(3,1,1)}(T)| = 4$, and indeed T has 4 subtrees on 3 vertices: the three paths $\ell_0 v_0 v_1$, $v_0 v_1 \ell_1$, $v_0 v_1 \ell_1'$, and the claw $\ell_1 v_1 \ell_1'$.

(v) $|c_{(4,1)}(T)| = 3$, matching the 3 leaves of T.

(vi) The path sequence is $(\pi_1, \pi_2, \pi_3) = (4,4,2)$ (counting the four single edges, the four 2-edge paths, and the two 3-edge paths $\ell_0 v_0 v_1 \ell_1$ and $\ell_0 v_0 v_1 \ell_1'$); the degree sequence is $(\delta_1, \delta_2, \delta_3) = (3,1,1)$. Both are easily checked to be consistent with the data above.

6.2 Example: reconstructing a 9-vertex caterpillar

Consider the caterpillar T with leaf-number sequence $(f_0, f_1, f_2) = (1,2,3)$, with spine v_0, v_1, v_2. This caterpillar has $n = 3 + (1+2+3) = 9$ vertices, 4 leaf-numbers distinct and positive, so Theorem 5.1 applies. We demonstrate the four-step reconstruction procedure on the data $\{c_\lambda(T)\}$, without referring to the construction of T.



By direct computation (or via the SageMath code of Section 8), the singleton-free coefficients of $\mathbf{X}_T$ are:

$$c_{(4,3,2)}(T) = +1, \qquad c_{(5,4)}(T) = -1, \qquad c_{(7,2)}(T) = -1, \qquad c_{(9)}(T) = +1.$$

We now run the procedure of the proof of Theorem 5.1.

Step 1. *Identify the singleton-free length classes.* The maximum length of a partition λ with $c_\lambda(T) \neq 0$ and singleton-free λ is 3, attained uniquely by $\lambda = (4, 3, 2)$. Hence $s + 1 = 3$ (so $s = 2$) and $\lambda = (4, 3, 2)$.

Step 2. *Read off the multiset of leaf numbers.* The parts of λ are $4, 3, 2$, so subtracting 1 gives the multiset $\{f_0, f_1, f_2\} = \{3, 2, 1\}$. All three are positive and distinct, confirming that Theorem 5.1 applies.

Step 3. *Find the partitions μ_i.* The singleton-free partitions of length $s = 2$ with non-zero coefficient are $(5, 4)$ and $(7, 2)$. These must be $\{\mu_1, \mu_2\}$ in some order.

Step 4. *Identify the merged pair for $\mu_1 = (5, 4)$.* Comparing $\mu_1 = (5, 4)$ to $\lambda = (4, 3, 2)$, the multiset of parts of λ absent from μ_1 is $\{3, 2\}$. Hence the endpoints of one spine edge have leaf numbers $\{3 - 1, 2 - 1\} = \{2, 1\}$.

Step 5. *Identify the merged pair for $\mu_2 = (7, 2)$.* Comparing $\mu_2 = (7, 2)$ to $\lambda = (4, 3, 2)$, the missing parts are $\{4, 3\}$, so the endpoints of the other spine edge have leaf numbers $\{3, 2\}$.

Step 6. *Build the auxiliary path graph.* The auxiliary graph H has vertex set $\{1, 2, 3\}$ and edges $\{1, 2\}$ and $\{2, 3\}$, which is the path 1–2–3.

Step 7. *Read off the leaf-number sequence.* The path 1–2–3 has the unique walk $(1, 2, 3)$ up to reversal. Hence $(f_0, f_1, f_2) = (1, 2, 3)$ up to reversal.

This recovers T up to isomorphism, as predicted by Theorem 5.1.

7 A conjecture and partial progress

Theorem 5.1 reconstructs caterpillars whose leaf numbers are strictly positive and pairwise distinct, and Remark 5.2 sketches a slight relaxation. The paper [6] also notes (citing [7]) that *symmetric* caterpillars are reconstructible. Combining these two pieces of evidence with Tan's computational confirmation of the Stanley conjecture for trees up to 23 vertices [10], we are led to the following.

Conjecture 7.1 (Caterpillar reconstruction)**.** *Every caterpillar is determined up to isomorphism by its chromatic symmetric function.*

Evidence for Conjecture 7.1.

1. Theorem 5.1 settles the case of pairwise distinct positive leaf numbers, and Remark 5.2 settles the case where vertices of equal leaf number form a contiguous block.

2. Symmetric caterpillars are reconstructible from $\mathbf{X}_T$ by results of Gordon–McDonnell [4] and Morin [7, Thm. 4.3.1].

3. Tan [10] verified Stanley's conjecture for all trees on at most 23 vertices. In particular, no two non-isomorphic caterpillars on ≤ 23 vertices share a chromatic symmetric function.

4. By Eisenstat–Gordon [2], there exist infinite families of caterpillar pairs with the same subtree polynomial; Theorem 1 of [6] shows that the chromatic symmetric function is strictly stronger than the subtree polynomial. Concretely, Figure 2 of [6] exhibits the smallest such pair, a pair of caterpillars on 11 vertices, and one of the authors verifies directly that $c_{(7,2,2)}(T_1) \neq c_{(7,2,2)}(T_2)$, so the chromatic symmetric functions distinguish them.

5. Our own SageMath implementation (Section 8) extends this empirical evidence by enumerating all non-isomorphic caterpillars on $n \leq 12$ vertices (a total of 271 caterpillars at $n = 12$) and verifying that no two of them share a chromatic symmetric function.

A partial result. We can establish a small but new step toward Conjecture 7.1: even without distinctness, the chromatic symmetric function determines the multiset of leaf numbers of any caterpillar with positive leaf numbers.

Proposition 7.2. *Let T be a caterpillar with spine $v_0, \ldots, v_s$ and leaf numbers $f_0, \ldots, f_s$, and assume every $f_i \geq 1$. Then the multiset $\{f_0, f_1, \ldots, f_s\}$ is determined by $\mathbf{X}_T$.*

Proof. By Proposition 4.1(vi), $\mathbf{X}_T$ determines whether T is a caterpillar (via the equality $\operatorname{diam}(T) - 1 = n - \delta_1(T)$ from [6, Prop. 10]) and, if so, the values of $s + 1$ and n. Step 1 of the proof of Theorem 5.1 did not use distinctness: edge sets with singleton-free type are exactly the supersets of the leaf-edge set L. Hence the unique singleton-free partition of maximum length $s + 1$ with non-zero coefficient $c_\lambda(T)$ is $\lambda = \operatorname{type}(L)$, whose parts are exactly $f_i + 1$ for $i = 0, 1, \ldots, s$. In conclusion, subtracting 1 from each part of λ recovers the multiset $\{f_0, f_1, \ldots, f_s\}$ of leaf numbers, completing the proof. □

Proposition 7.2 reduces Conjecture 7.1 (in the case of positive leaf numbers) to the question of recovering the cyclic order of the multiset of leaf numbers along the spine, given that we know the multiset. In the distinct-leaf-numbers case (Theorem 5.1), this is achieved by examining the partitions μ_i. The natural next step would be to show that even when leaf numbers repeat, the higher-order invariants $\operatorname{type}(L \cup B)$ for $|B| \geq 2$ disambiguate the order. This requires controlling cancellations in $c_\mu(T)$ when several edge sets share a type, which appears to be subtle but tractable.

8 Computational aspects

The accompanying SageMath module `csf_trees.sage` implements the constructions and reconstructions of this report directly. Its main components are:

- a routine `chromatic_symmetric_function(G)` that computes $\mathbf{X}_G$ for any Sage graph G as a genuine element of `SymmetricFunctions(QQ).power()`, by enumerating all $2^{|E(G)|}$ edge subsets and applying Stanley's expansion (1);

- a routine `caterpillar(leaf_numbers)` building the caterpillar with a given leaf-number sequence;

- a routine `reconstruct_caterpillar(coeffs)` that, given the coefficient dictionary $\{c_\lambda(T)\}$, performs the four-step procedure of Section 5 to recover the caterpillar (under the hypotheses of Theorem 5.1);

- a brute-force search across all non-isomorphic caterpillars with $n \leq 12$ vertices, comparing chromatic symmetric functions in pairs.

The implementation serves three distinct purposes that go beyond mere verification. First, it provides an executable, machine-checkable certificate of the proof of Theorem 5.1: the procedure `reconstruct_caterpillar` succeeds on every valid input and fails predictably (with a diagnostic) when the distinctness hypothesis is violated, confirming that the four-step argument is constructive. Second, the brute-force search supplies new empirical support for Conjecture 7.1 *beyond* the f_i-distinct case handled by Theorem 5.1. In particular, the search includes caterpillars with repeated and zero leaf numbers (cases not covered by Theorem 5.1 or Remark 5.2), and finds no collisions through $n = 12$. This complements Tan's verification of Stanley's full tree conjecture [10] by giving caterpillar-specific counts ($1, 2, 5, 9, 19, 35, 71, 135, 271$ caterpillars for $n = 4, \ldots, 12$, respectively), all of which have distinct chromatic symmetric functions. Third, the symmetric-function output makes further algebraic exploration straightforward (e.g. basis change to the Schur or homogeneous bases, scalar products, and so on) using SageMath's built-in tools.

The README accompanying the code describes how to run all three demos and reproduce every numerical claim made in this report.

A Response to peer review

This appendix briefly summarizes the changes made between the April 29 draft and the final May 13 submission, in response to the peer review by James Gillin (May 6).

Item 1: "Complete isomorphism invariant" clarification (Section 1).
Added a parenthetical immediately after the first use of the phrase "not a complete isomorphism invariant" to explain the term, and reinforced the definition again when stating the tree-restricted version of the question.

Item 2: Transition to equation (2) (Section 3).
Inserted an explanatory passage immediately before equation (2) that explicitly states why the sign $(-1)^{|A|}$ is constant on edge subsets of a fixed type, namely that on trees the length of the type alone determines $|A|$.

Item 3: Missing verb in proof of Theorem 5.1.
Reworded the opening sentence of the proof so that the leaf-edge set is properly introduced ("Let $L \subseteq E(T)$ denote the set of leaf edges of $T, \ldots$"), repairing the ungrammatical fragment in the draft.

Item 4: Step 1 implication that $L \subseteq A$.
Expanded the explanation of why singleton-free type forces $L \subseteq A$, adding a clarifying sentence pointing out that a leaf can only avoid being an isolated singleton component if its unique incident edge is included in A.

Item 5: Density of the table in Section 6.1.
Added a short reading guide immediately before the table explaining the row organization (rows by $|A|$, grouping by type) and how the signed contribution to $c_\lambda(T)$ is computed from each row.

Item 6: Conclusion of Proposition 7.2.
Rewrote the final sentence of the proof to state the conclusion explicitly: subtracting 1 from each part of λ produces the multiset $\{f_0, \ldots, f_s\}$.

Item 7: Section 8 insight beyond verification.

Replaced the brief description of the SageMath module with a substantive paragraph explaining the three distinct purposes of the implementation: (a) a constructive certificate of Theorem 5.1; (b) new empirical evidence for Conjecture 7.1 beyond the cases the theorem covers; and (c) algebraic extensibility via SageMath's symmetric-function infrastructure. Included the explicit caterpillar counts for $n = 4, \ldots, 12$.

In addition to the items above, the date and author block were updated to reflect the final-submission status, and the introduction's roadmap paragraph was extended to reference this appendix.

References

[1] B. Bollobás, *Modern Graph Theory*, Graduate Texts in Mathematics 184, Springer-Verlag, New York, 1998.

[2] D. Eisenstat and G. Gordon, *Non-isomorphic caterpillars with identical subtree data*, Discrete Math. **306** (2006), 827–830.

[3] J. Fougere, *On symmetric chromatic polynomials of trees*, undergraduate thesis, Dartmouth College, 2003.

[4] G. Gordon and E. McDonnell, *Trees with the same degree sequence and path number*, Discrete Math. **147** (1995), 297–300.

[5] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford University Press, New York, 1995.

[6] J. L. Martin, M. Morin, and J. D. Wagner, *On distinguishing trees by their chromatic symmetric functions*, J. Combin. Theory Ser. A **115** (2008), 237–253.

[7] M. Morin, *Caterpillars, Ribbons, and the Chromatic Symmetric Function*, M.S. thesis, University of British Columbia, 2005.

[8] R. P. Stanley, *A symmetric function generalization of the chromatic polynomial of a graph*, Adv. Math. **111** (1995), 166–194.

[9] R. P. Stanley, *Enumerative Combinatorics, Vol. 2*, Cambridge University Press, Cambridge, 1999.

[10] L.-Y. Tan, *Computational verification that the chromatic symmetric function determines up to isomorphism trees of order at most* 23, available at `http://www.cs.wustl.edu/~lt1/csf.html`, retrieved 2007.